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PI Name:	Fornace, Albert M.D.		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Key Personnel Changes/Previous PI:	July 2022 report: Shubhankar Suman added as Co-Investigator in the year-6 of this NASA Specialized Center of Research (NSCOR).		
COI Name (Institution):	Brenner, David Ph.D. (Columbia University) Meltzer, Paul M.D., Ph.D. (National Cancer Institute, NIH) Shay, Jerry Ph.D. (University of Texas Southwestern Medical Center at Dallas) Suman, Shubhankar Ph.D. (Georgetown University)		
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Task Description:	Considering the high spontaneous incidence of gastrointestinal (GI) and lung cancer, an even modest increase by space radiation exposure could have a significant effect on astronauts' health risk estimates during and after long-duration space flights. However, there is substantial uncertainty for GI and lung cancer risk estimation from space radiation due to the lack of in vivo human data. Our overall goal was to specifically investigate cellular and molecular hypothesis-driven mechanisms in relevant mouse models that contribute to carcinogenic risk estimates of GI (GU and UTSW) and lung cancer (UTSW) after exposure to space radiation beams prioritized by NASA's human research program (HRP). Specifically, we determined the incidence, tumor frequency, and grade as well as molecular perturbations in the target cells associated with GI and lung tumorigenesis through monitoring of changes in the tumor number, histology, and gene/protein expression of the utilized model systems after exposure to space-radiation type beams. The overall hypothesis is that radiation that is prevalent in space is more carcinogenic due to important qualitative differences from ? rays where human exposure data are available. Molecular events triggered by space radiation will be investigated in-depth, and the mechanistic basis for persistent aberrant cell-signaling after space radiation will be dissected. Both quantitative and qualitative murine tumor data will contribute to the modeling of human risks for GI and lung cancer by space travel. We plan to test our hypothesis by pursuing the following specific aims: Aim 1. Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2. Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3. Quantitatively assess lung tumorigenesis and test potential countermeasures in lung cancer susceptible mouse model. Aim 4. Development of mathematical modeling for cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Considering the high frequency of gastrointestinal and lung cancer in the US population, an even small increase by space radiation could have a major impact on risk estimates and the planning of future manned space missions. In addition to risk estimation, studies on the persistence of oxidative and inflammatory stress after simulated space radiation and its heavy-ion components and their role in driving gastrointestinal and lung tumorigenesis may provide insight into mutagenic processes affecting genome integrity and carcinogenesis. The significance and deliverables of this project are to improve the estimates of GI and lung cancer risk in astronauts and to identify plausible targets for the future development of mitigation strategies.
	Pointwise progress made for each aim/project is summarized below: GI-tumorigenesis study (Project 1, Georgetown University): In the last eight years of NASA Specialized Center of Research (NSCOR) support, we have successfully applied mouse model approaches to acquire quantitative and qualitative gastrointestinal (GI) tumorigenesis data to address prevailing uncertainties in risk modeling. Our overall goal is to determine the relative biological effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation. The risk estimates can then be extrapolated to space radiation using a RBE scaling factor. Pointwise details of progress made for GI tumorigenesis study are provided below: 1. Using Apc1638N/+ mice, we reported that the relative RBE for radiation-induced (IR)-induced gastrointestinal (GI) tumorigenesis is related to ion energy, linear energy transfer (LET), and gender. Male and female mice were exposed to sham or 5 to 200 cGy doses of ??-rays, 4He (250 MeV/n), 12C (290 MeV/n), 16O (325 MeV/n), 28Si (300 MeV/n), and 56Fe (1000 MeV/n). Calculation of RBE for intestinal and colon tumorigenesis showed the highest value with 28Si, and lower doses showed greater RBE relative to higher doses. 2. We reported downregulation of RXRa, an Apc-independent factor involved in targeting ß-catenin to the ubiquitin-proteasome pathway (UPP) for degradation after heavy-ion exposure. Decreased expression of RXRa in tumors as well as in adjacent normal epithelium was indicative of perturbations in ß-catenin proteasomal-targeting machinery, which suggests the plausibility of RXRa agonists as a potential countermeasure against heavy-ion-induced GI tumorigenesis. 3. We compared GI tumorigenesis in Apc1638N/+ mice after exposure to heavy ions at a high (50 cGy/min) and relatively low (0.33 cGy/min) dose rate. In both male and female mice, GI tumor number and size were not significantly different after high and low dose rate exposures. This study suggested that the carcinogenic potential of energetic heavy ions is independent of the dose rate. 4. Significantly higher intestinal tumor number and grade, along with decreased cell differentiation, were observed after acute radiation relative to fractionated radiation. Acute protons induced upregulation of ß-catenin and Akt pathways with increased proliferative marker phospho-histone H3. Increased DNA damage along with decreased DNA repair factors involved in mismatch repair and nonhomologous end joining were also observed after exposure to acute protons. 5. A comparison of tumor data from simplified-mixed-field (Smf)-galactic cosmic radiation (GCR) and equivalent doses of heavy ions revealed an association between higher GI tumorigenesis, where the dose received from heavy ions probably contributed to much of the total GI tumorigenic effect observed after Smf-GCR. This study provided the first experimental evidence that cancer risk after GCR exposure may largely depend on doses received from constituent heavy ions. 6. Aspirin (an inhibitor of COX), was evaluated as a potential countermeasure against 28Si-induced GI tumorigenesis in Apc1638N/+. Aspirin led to a significant reduction in PGE2 in a dose-dependent manner but did not reduce 28Si-induced GI tumorigenesis. 7. Metformin is a known suppressor of IGF1-mTOR signaling via activation of AMP-activated protein kinase (AMPK), we tested its potential GI-cancer preventive effects against heavy ion-induced GI tumorigenesis. We noted that long-term continuous intake of metformin confers significant protection against heavy-ion radiation-induced GI tumorigenesis. 8. We assessed radiation quality effects on colonic inflammation, colon cancer incidence, and associated signaling events in Il10-/- mice. IR-induced colitis and colitis-associated cancer (CAC) incidence in Il10-/- mice depend on radiation

Task Progress:

quality and display co-activation of ß-catenin and NF-?B signaling. Regardless of the Apc gene status of a mouse model, heavy-ion IR-induced GI tumorigenesis displays co-activation of oncogenic and pro-inflammatory signaling pathways.

9. Male Apc1638N/+ mice were exposed to 5, 10, and 50 cGy ?? rays or 28Si (69 keV/µm) radiation, and gastric tumorigenesis was assessed at 150 days post-exposure relative to the control group. Animals exposed to 28Si radiation had the most gastric tumors and carcinomas, followed by ?-ray-exposed and control groups. Both 28Si and ?-ray exposures showed a dose-dependent increase in gastric tumor and carcinoma development at the dose range of 5 to 50 cGy. At all doses, the radiation-quality-dependent increase in gastric tumorigenesis was evident. Our findings suggest that exposure to heavy ions is more likely to cause gastric cancer than low-LET radiation.

Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation (Project-2, Georgetown University):

Proliferative gastrointestinal (GI) tissue is radiation-sensitive, and heavy-ion space radiation with its high-linear energy transfer (high-LET) and higher damaging potential than low-LET ?-rays is predicted to compromise astronauts' GI function. However, much uncertainty remains in our understanding of how heavy ions affect coordinated epithelial cell migration and extrusion, which are essential for GI homeostasis. The overall goal of this project is to understand the long-term effects of space radiation on normal gastrointestinal (GI) cells including stem cells.

Pointwise details of progress made for persistent effects of space radiation on GI tissues are provided below:

1. We investigated the long-term effects of 56Fe radiation on adipokines and the insulin-like growth factor 1 (IGF1) signaling axis in the mouse intestine and colon. Irradiation increased leptin and IGF1 levels in serum and IGF1R and leptin receptor expression in tissues. Additionally, upregulated Jak2/Stat3 pathway and cell proliferation were also noted. Therefore, our data demonstrated chronic pathophysiologic and endocrine alterations in GI tissue after space radiation exposure.

2. We demonstrated that heavy-ion exposure resulted in persistently delayed intestinal epithelial cell (IEC) migration involving chronic sublethal genotoxic and oncogenic stress-induced altered cytoskeletal dynamics, which were seen even a year later after IR exposure.

3. 56Fe radiation triggered a time-dependent increase in ?H2AX foci and senescent cells but without a noticeable increase in apoptosis. Some senescent cells acquired the senescence-associated secretory phenotype (SASP), and this was accompanied by increased intestinal epithelial cell (IEC) proliferation. Additionally, using the Lgr5-EGFP-IRES-creERT mice model, we observed increased ROS and ongoing DNA damage by staining for ?H2AX, and 53BP1, along with the accumulation of senescence markers in intestinal stem cells. Results also showed increased acquisition of SASP in senescent intestinal stem cells (ISCs) after heavy-ion exposure relative to ?-rays. Collectively, our data indicate that heavy-ion-induced chronic stress and ongoing DNA damage promoted SASP in a fraction of the ISCs.

4. Exposure to 56Fe resulted in significantly higher levels of DNA base lesions in the GI tissue, relative to control and ?-radiation. Both base excision repair (BER) and nucleotide excision repair (NER) pathways were downregulated. Moreover, DNA damage response (DDR) alterations could result in both somatic gene mutations as well as activation of the pro-inflammatory, pro-oncogenic SASP signaling known to accelerate adenoma-to-carcinoma progression during radiation-induced GI cancer development.

5. In addition to GI tissues, differential accumulation of senescence and SASP cells was observed as part of the persistent IR-induced alterations in mouse bone marrow (BM). A remarkable increase in p16-positive cells indicated a persistent increase in cell senescence; whereas, an increased RANKL/OPG ratio, reductions in the number of osteoblast progenitor cells, and osteocalcin provided clear evidence that exposure to both proton and 56Fe-ions promotes pro-osteoclastogenic activity in BM.

6. Senescent cells can be removed intrinsically by immune cells. 28Si exposure decreased the NK cell population in the LP, which suggests inefficient immune surveillance of senescent cells that could contribute to the persistent accumulations of senescent and SASP cells after heavy ion exposure.

7. Exposure to a low dose of ?-rays were associated with upregulation of nicotinamide adenine dinucleotide phosphate (NADPH) oxidase and its modifiers, along with increased Ras/p38/Gata6 signaling in the colon. When considered along with oxidative damage and proliferative markers, our observations suggest that the NADPH oxidase pathway could be playing a critical role in propagating long-term oxidative stress after radiation.

Lung tumorigenesis and countermeasure studies (UTSW) (Project-3):

1. Developed an isogenic series of human colon and lung normal, premalignant and cancer cell lines to investigate carcinogenesis after low and high LET irradiation. 2. Determined that 3D culture of human normal colonocytes can be used to investigate radiation-induced transformation.

3. Quantitatively assessed lung and colon tumorigenesis in mouse models of radiation-associated cancer susceptibility. Determined that the low threshold for polyp formation in the colon cancer susceptible mouse model was between 5 and 10cGy of HZE silicon particle irradiation.

4. To simulate the total dose expected by astronauts on a round trip to Mars, we utilized a simplified 3-beam GCRsim fast-switching approach in mice using a total of 30 cGy or less. 30 cGy single-dose of protons and a single dose of 5 cGy silicon did not increase carcinogenesis.

5. This 3-beam GCR simulation used 20cGy proton, 5cGy helium, and 5cGy silicon which resulted in an increase in plasma lipid peroxidation (a biomarker of increased persistent inflammation). We also observed an increase in premalignant lesions and advanced cancers (~2-fold). However, if we reduced the amount of silicon to 2 cGy the increased cancer risk was decreased back to unirradiated control mice. If we reversed the order of ions so that silicon was provided first followed by helium and then protons, we did not observe the increases in cancer risk. We concluded that the order of GCR irradiation matters.

6. Determined that aspirin does not protect against space radiation-associated cancers.

7. Demonstrated that Nrf2 activation by the synthetic triterpenoid, CDDO (an FDA approved oral drug to treat patients

with Friedreich's Ataxia), protects colonic epithelial cells against IR-induced transformation in part by enhancing repair of DNA damage. CDDO prevents human cell transformation using both low and high LET irradiation. CDDO protects specific brain functions after GCR simulations and protects normal lung, colon, and breast epithelial cells but not cancer cells from irradiation. CDDO increases mouse survival after 10Gy total body ionizing radiation.

8. Metformin pre-administration acts as a radioprotector through AMPK phosphorylation, increasing OGG1 and decreasing cleaved PARP expression. Altogether, we interpret these results to suggest that metformin is a potential oral available and safe GCR radioprotector and has the potential to lower the risk of cancer initiation/promotion in astronauts.

Cancer risk modeling (Columbia University) (Project-4):

Long-duration space exploration missions, such as a journey to Mars, pose significant health risks to astronauts due to exposure to a unique mixture of sparsely ionizing and densely ionizing radiations present in space. The biological effects of this mixture can be divided into two categories: targeted effects (TE) and non-targeted effects (NTE). The combination of both TE and NTE can result in non-linear dose-response shapes for space radiations, particularly in the dose range relevant for space exploration missions. To predict the likelihood of developing cancer and other diseases following space travel, it is important to use radiation dose-response models that incorporate both TE and NTE components. In this project, such a model was developed and applied to data on intestinal and gastric carcinogenesis in Apc1638N/+ tumor-prone mice and are described below:

1. Dose-response and dose rate effect analyses: Modeling of our intestinal carcinogenesis data using the TE+NTE formalism suggested considerable NTE involvement at low doses that are relevant for space travel. The NTE contribution was predicted to be particularly strong for the most densely ionizing tested ions Si and Fe. Model-based estimates for the dose-rate effect on heavy ion-induced carcinogenesis at doses/dose rates expected during a Mars mission show only a small and not statistically-significant dose-rate effect. Consequently, modeling analysis suggests that heavy ion carcinogenesis estimates from moderate/high dose-rate experimental data may be applicable to doses/dose rates relevant for space exploration.

2. Metrics for radiation quality comparisons: Reliable methods are needed for scaling low-LET to high-LET radiation risks for humans, based on animal or in vitro studies comparing these radiations. A proposed new metric, radiation effects ratio (RER), compares the effects of two radiations at the same dose of interest (vertical scaling). In contrast, the traditional RBE metric relies on horizontal comparisons of dose responses for the compared radiations, for which more extensive information at several doses is generally needed. RER can be used without the need for detailed information about dose-response shapes for compared radiations and allows animal carcinogenesis experiments to be simplified by reducing the number of tested radiation doses. For simple linear dose-effect relationships, RBE = RER. However, for more complex dose-effect relationships, such as those with NTE, RER can be different from RBE.

3. Analyses of radiation mixture effects: Based on the modeled responses for each radiation type, the tumor yield for a Mars-mission-relevant mixture of these radiations was predicted using the recently developed incremental effect additivity (IEA) synergy theory. The proposed modeling approach can enhance current knowledge about the quantification of space radiation quality effects, dose-response shapes, and ultimately the health risks for astronauts.

4. Microdosimetric quality factor calculations: The microdosimetric approach, using measured or calculated distributions of microdosimetric energy deposition together with empirical biological weighting functions, is conceptually and practically simpler than the alternative fluence-based risk cross-section approach. The microdosimetric approach is practical and can allow continuous readout of biologically effective doses during spaceflight in conjunction with in-situ measurements of microdosimetric spectra.

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